



Laxmi Charitable Trust's
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Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Practical – 8

Roll No. :	Name:
Class :	Date/Time of Submission: 11 -08-2026

Practical No. 8 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

a. Write a program to accept and pass radius to a function that returns area and circumference (using tuple).

Code:


```
prac8.py - C:/Users/mvluc/Desktop/prac8.py (3.14.6)
File Edit Format Run Options Window Help

#Practical 8b: Write a program to perform basic file operations on text files
#and binary files.

# Practical 8 b-1) Text File Operations

#Write to a text file
with open("example.txt", "w") as f:
    #The with statement ensures the file will automatically close after write/read
    f.write("Hello Students!\n Welcome to Python File Handling.")

# Read from text file
with open("example.txt", "r") as f:
    content = f.read()
    print("Text File contents:\n", content)
|
```

Output:

```
>>> ===== RESTART: C:/Users/mvluc/
Text File contents:
Hello Students!
Welcome to Python File Handling.
>>>
```

b)

Code:

```
practopy C:/Users/mvive/Desktop/practopy (3/17/20)
File Edit Format Run Options Window Help
#Practical 8b:ii) Write a program to perform basic file operations on binary files

#Binary File Operations

#Write binary data(bytes) to a File
with open("b_sample.bin", "wb") as bf:
    data = b"This is binary data"
    bf.write(data)

# Read binary data from the file
with open("b_sample.bin", "rb") as bf:
    b_content = bf.read()
    print("Binary File contents:\n", b_content)
|
```

Output:

```
>>> ===== RESTART: C:/Users/m
Binary File contents:
b'This is binary data'
>>> |
```

c. Write a Python program to read last n lines of a file.
Code:

```
*prac8.py - C:/Users/mvluc/Desktop/prac8.py (3.14.6)*
File Edit Format Run Options Window Help
def read_last_n_lines(filename, n):
    with open(filename, "r") as f:
        lines = f.readlines() # Reads all lines from the file into a list called
                               # lines. So - lines = ["Line 1\n", "Line 2\n", "Line 3\n"]
        return lines[-n:]    # Get last n lines. Negative indexing (-n:)
                               # is used to slice the list from the end.
                               # Example: ["Line 2\n", "Line 3\n"]

# File name and n value
filename = "example.txt"
n = int(input("Enter number of last lines to read: "))

# Read and print lines
# Executes the function and stores the result in last_lines
last_lines = read_last_n_lines(filename, n)

print("\nLast", n, "Lines:")
for line in last_lines:
    print(line.strip()) # Removes extra spaces and newline characters (\n)
                       # Prints each line cleanly without blank lines.
```

Output:

```
>>> ===== RESTART: C:/Users/mvluc/Desktop/prac8
Enter number of last lines to read: 2

Last 2 Lines:
Hello Students!
Welcome to Python File Handling.
>>>
```

Q. 2. What is file? What are its operating modes?

Q. 3. Write short note on file object attributes and file methods.